Portland State University

DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING

ECE332

LAB 4: PATCH ANTENNA

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Contents

1 Introduction

This lab builds and characterizes a half-wave rectangular patch antenna on single-sided copper-clad FR4, tuned for the 2.4 GHz WiFi band ($f_c \approx 2.443$ GHz, the geometric mean of the 2.400–2.487 GHz channel range). The patch is fed at the edge by a 50 Ω microstrip transmission line made from copper tape, and a side-launch SMA connector ties the transmission line to the VNA. The patch length L sets the resonant frequency through the half-wavelength-in-dielectric condition $f_c = c/(2L\sqrt{\varepsilon_r})$. The width W controls the input impedance and the radiation pattern, and the inset distance R tunes the feed-point impedance down toward 50 Ω.

After construction we measure S_{11} and S_{22} on a VNA to verify the resonance lands inside the WiFi band, trim the patch length if needed, then cut an inset feed of length R to drop the input impedance to 50 Ω. The second half of the lab uses both antennas at once to measure S_{21} and S_{12} versus orientation angle, producing an azimuth pattern of the radiation lobe. A cross-polarization test (90° rotation of one antenna) verifies that patch antennas are linearly polarized in the far field.

2 Antenna Design Calculations

Patch length L versus ε_r

The center frequency of the patch is set by the half-wavelength-in-dielectric condition,

$$f_c \approx \frac{c}{2L\sqrt{\varepsilon_r}} \quad \Longrightarrow \quad L \approx \frac{c}{2f_c\sqrt{\varepsilon_r}}.$$

With $c = 3 \times 10^8$ m/s and $f_c = 2.443$ GHz, the numerator is $c/(2f_c) = 61.40$ mm.

ε_r	L (mm)
4.0	30.70
4.1	30.32
4.2	29.96
4.3	29.61
4.4	29.27

About the spread: a 10% sweep in ε_r (4.0 to 4.4) only moves L by about 1.43 mm, which is roughly 4.7% of L. The length scales as $1/\sqrt{\varepsilon_r}$, so the sensitivity is half the relative error in ε_r. For example a ±0.1 swing around $\varepsilon_r = 4.2$ shifts L by about ±0.36 mm, which is right at the edge of what an Xacto knife can resolve. So basically picking the wrong ε_r inside the FR4 spec range won't kill the resonance, it just nudges f_c by a few tens of MHz and we trim by hand on the VNA anyway.

Design choice: use $\varepsilon_r = 4.2$ for the initial cut, so $L =$ *29.96 mm* as the starting length.

Patch width W

The constraint from the lab is $W \geq L$, with larger W dropping the input impedance and broadening the pattern. A square patch ($W = L \approx 30$ mm) sits around 300 Ω, so we picked $W = 38$ mm.

About 25% wider than L, which trims the edge impedance from $\sim 300\ \Omega$ down toward $\sim 200\ \Omega$ and leaves enough copper area on each side of the transmission line to cut the inset feed without nicking the radiating edge. W runs along the long edge of the FR4 board so the patch and feed line fit comfortably without crowding the SMA connector.

50 Ω transmission line width

For a microstrip line on FR4 with substrate height $h = 1.6$ mm and $\varepsilon_r = 4.2$, the 50 Ω trace width comes from the Hammerstad/Wheeler synthesis formula. For $Z_0 = 50\ \Omega$ on standard FR4 the ratio w/h lands at about 1.92, so

$$w \approx 1.92 \cdot h = 1.92 \cdot 1.6 \text{ mm} \approx 3.05 \text{ mm}.$$

We rounded to $w = 3.0$ mm for the cut, which puts Z_0 within about 1 Ω of target, well inside what the VNA-trimming step can clean up.

Wavelength and far-field boundary

The free-space wavelength at $f_c = 2.443$ GHz is

$$\lambda_0 = \frac{c}{f_c} = \frac{3 \times 10^8}{2.443 \times 10^9} \approx 12.28 \text{ cm}.$$

For an aperture antenna with largest dimension D, the far-field (Fraunhofer) region begins at

$$r_{ff} = \frac{2D^2}{\lambda_0}.$$

With $D = W = 38$ mm, $r_{ff} \approx 2(0.038)^2/0.1228 \approx 2.35$ cm. So any cable-length separation (typically ≥ 30 cm with the lab's blue SMAs) is comfortably in the far-field, and the azimuth pattern we measure will be the true radiation pattern rather than near-field coupling.

3 VNA Setup

The VNA was configured per the lab procedure with a 4-way quadrant split: top-left S_{11} Log-Mag, top-right S_{11} Smith (R+jX), bottom-left S_{22} Log-Mag, bottom-right S_{22} Smith (R+jX). Start frequency 1.5 GHz, stop frequency 2.7 GHz. The eCal module was used for calibration on both ports through the blue SMA cables. After the resonance was found in the WiFi band, the VNA was recalibrated over the narrower 2.35–2.55 GHz span.

4 Build the Antennas

Both antennas were built on single-sided copper-clad FR4. The patch and 50 Ω transmission line were cut from copper tape using an Xacto knife, with the transmission line meeting the patch at the center of the W-edge and running to the board edge where the side-launch SMA connector was soldered. The SMA center pin attaches to the transmission line and the body pins to the ground plane on the bare side of the board.

Photographs

Insert photo of Bao's antenna with L, W, R, and transmission line dimensions marked in Sharpie.

Insert photo of Nick's antenna with L, W, R, and transmission line dimensions marked in Sharpie.

As-built dimensions

	Bao's antenna	Nick's antenna
L initial cut (mm)	29.96	29.96
L after trim (mm)	*TBD*	*TBD*
W (mm)	38.0	38.0
Transmission line width (mm)	3.0	3.0
Transmission line length (mm)	*TBD*	*TBD*
Inset distance R (mm)	*TBD*	*TBD*

5 Initial Measurements and Impedance Matching

Initial resonance and trimming

With both antennas connected, S_{11} (Bao) and S_{22} (Nick) were swept from 1.5–2.7 GHz to find the initial resonance. The frequency with the lowest return-loss magnitude was treated as the operating frequency.

	Bao (S_{11})	Nick (S_{22})
Initial f_{ant} (GHz)	*TBD*	*TBD*
Trim direction (cut/extend L)	*TBD*	*TBD*
Post-trim f_{ant} (GHz)	*TBD*	*TBD*

Describe what the initial resonance looked like, why it was off from 2.443 GHz, and how much length was remove

Input impedance measurement

At the post-trim operating frequency, the input impedance was read from the Smith chart marker.

	Bao's antenna	Nick's antenna
Z_{IN_0} from Smith chart (Ω)	*TBD*	*TBD*
Return loss at f_c (dB)	*TBD*	*TBD*

Inset feed calculation R

Moving the feed inward by distance R scales the input impedance as

$$Z_{IN\_R} = Z_{IN\_0} \cos^2\left(\frac{\pi R}{L}\right) \quad \Longrightarrow \quad R = \frac{L}{\pi} \cos^{-1} \sqrt{\frac{Z_{IN\_R}}{Z_{IN\_0}}}$$

where $Z_{IN\_R} = 50\ \Omega$ is the target and Z_{IN_0} is the measured edge-fed impedance.

Calculated R (plug in measured L and Z_{IN_0}):

- Bao's antenna: $L =$ *TBD*, $Z_{IN\_0} =$ *TBD* $\Omega \Rightarrow R =$ *TBD mm*
- Nick's antenna: $L =$ *TBD*, $Z_{IN\_0} =$ *TBD* $\Omega \Rightarrow R =$ *TBD mm*

After inset cuts

Per the lab procedure, about half of the calculated R was cut from each side of the patch, then the VNA was monitored in Log-Mag mode and additional symmetric cuts made until return loss bottomed out.

	Bao's antenna (S_{11})	Nick's antenna (S_{22})
Final R (mm)	*TBD*	*TBD*
Return loss at f_c (dB)	*TBD*	*TBD*
Return loss at 2.400 GHz (dB)	*TBD*	*TBD*
Return loss at 2.487 GHz (dB)	*TBD*	*TBD*

Target: return loss below -20 dB at f_c, and at least -9.5 dB across the whole 2.400–2.487 GHz band.

VNA screen captures

Insert VNA screen capture showing S_{11} and S_{22} Log-Mag and Smith plots with both antennas connected, with th

6 More Measurements and Polarization Observations

Azimuth pattern

The Smith chart quadrants were reconfigured to show S_{12} and S_{21} Log-Mag instead. The two antennas were placed facing each other, separated by a fixed distance set by the VNA cable lengths. One antenna was held stationary; the other was rotated in 15° increments out to 90°. The aligned (0°) measurement was taken as the 0 dB baseline, and the values below are the drop from that baseline.

Angle (°)	S_{21} (dB)	S_{12} (dB)	Drop from 0 dB baseline (dB)
0	*TBD*	*TBD*	0.0
15	*TBD*	*TBD*	*TBD*
30	*TBD*	*TBD*	*TBD*
45	*TBD*	*TBD*	*TBD*
60	*TBD*	*TBD*	*TBD*
75	*TBD*	*TBD*	*TBD*
90	*TBD*	*TBD*	*TBD*

Antenna separation distance: *TBD cm*.

Insert polar (azimuth) plot from Excel showing the radiation pattern across ±90°. Use the 6+ data points above;

Cross-polarization test

With both antennas facing each other, one was tipped on its side so that its E-field axis was rotated 90° relative to the other. S_{21}/S_{12} was recorded, then compared to the value with the two antennas facing entirely away from each other.

Configuration	S_{21} (dB)	S_{12} (dB)
Both facing, one tipped 90° (cross-pol)	*TBD*	*TBD*
Both facing away from each other (back-to-back)	*TBD*	*TBD*

Discuss: is cross-pol lower than back-to-back, or higher? If lower, the patches are well-polarized and back-to-back

For the chosen separation, the far-field boundary is $r_{ff} = 2D^2/\lambda_0 \approx 2.35$ cm (using $D = W = 38$ mm and $\lambda_0 = 12.28$ cm at 2.443 GHz, derived above). Our antenna spacing of *TBD cm* is well past this threshold, so the link is operating in the far-field and the cross-pol result reflects true polarization mismatch rather than near-field coupling.

7 Discussion and Conclusion

Summarize: did the resonance land where the design calculation predicted (within a few percent)? How many tri